

General L^2 Poisson Concentration

A sharp uniform-factor answer to Remark 5 of arXiv:2411.16010

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23 July 2026

Abstract

Gómez, Kalaj, Melentijević, and Ramos ask whether their sharp concentration inequality for analytic Hardy functions has an analogue for Poisson extensions of arbitrary L^2 boundary data. We give an affirmative answer with multiplicative factor 2, prove that 2 is the best factor uniform in the hyperbolic area, and show that the analytic sharp factor 1 fails for every fixed positive area. The upper bound is the analytic/anti-analytic splitting. Sharpness is forced by normalized Poisson kernels whose centers approach the boundary.

Status. Candidate full result; verifier verdict: likely valid. Human review is recommended, especially for normalization and novelty. The result determines the best factor uniform over all $s > 0$, not the fixed- s optimal constant.

1 Source question

The source is Gómez–Kalaj–Melentijević–Ramos, *Uniform stability of concentration inequalities and applications*, arXiv:2411.16010 [1]. Their Theorem 1.4 contains the qualitative inequality

$$\int_{\Omega} |f(z)|^2 (1 - |z|^2)^{-1} dA(z) \leq \pi \log\left(1 + \frac{s}{\pi}\right) \|f\|_{H^2}^2, \quad f \in H^2(\mathbb{D}), \quad (1)$$

where

$$d\mu(z) = (1 - |z|^2)^{-2} dA(z), \quad \mu(\Omega) = s.$$

Remark 5 on pp. 41–42 then asks:

Can one obtain a concentration inequality for Poisson extensions of general L^2 functions on the boundary of the unit disk?

2 Normalization and proof intuition

Give $\mathbb{T}$ normalized measure

$$dm(e^{it}) = \frac{dt}{2\pi}.$$

If

$$h(e^{it}) = \sum_{n \in \mathbb{Z}} a_n e^{int} \in L^2(\mathbb{T}, dm),$$

its Poisson extension is

$$u(re^{it}) = P[h](re^{it}) = \sum_{n \in \mathbb{Z}} a_n r^{|n|} e^{int}.$$

Remark 5. As noted in the introduction, Theorem 1.4 may be interpreted as a (sharp stability version of) a novel concentration inequality for Poisson extensions of functions in H^2 . It is immediate from a change of variables that these results transfer immediately to the upper half plane.

In that regard, it is worth noting that all results in this manuscript yield sharp stability versions of concentration inequalities in *general* simply connected domains. Indeed, one simply uses a biholomorphism from the unit disk to any such domain, in the same vein as we did in order to transfer from the upper half space to the unit disk in Section 2.

In particular, by applying this procedure to Theorem 1.4 one would obtain a result on Poisson extensions to such domains. It is important to note that the weights yielded by that result are inherently connected to the biholomorphism used, and that as such the definition of Hardy spaces on the domain is also intimately related to the one for the unit disk.

The latter remark motivates the following question: can one obtain a concentration inequality for Poisson extensions of *general* L^2 functions on the boundary of the unit disk? This question seems to go partly beyond the scope of techniques used here: the functions under consideration are no longer analytic functions in the unit disk, but merely harmonic functions. Moreover, the crucial property used in [28,

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[24, 20, 18, 17, 19] that the functions under consideration are *log-subharmonic* is also not valid in general, and this problem hence seems to require new ideas. We wish to address this issue in a future work.

Figure 1: The complete source question, spanning pp. 41–42 of arXiv:2411.16010.

Proof intuition. The obstruction noted in the source is the failure of log-subharmonicity for a general harmonic function. There is no need to restore it. The positive and negative Fourier halves of u are separately analytic, so the source inequality applies to each half. Their interference costs at most a factor 2.

That loss is genuinely necessary in a uniform theorem. A normalized boundary Poisson kernel has nearly equal analytic and anti-analytic halves. Moving its center and a fixed pseudohyperbolic disk together toward the boundary makes the two halves reinforce. The pullback to a fixed Euclidean disk leaves one elementary angular integral, from which both the failure of factor 1 and the sharpness of factor 2 follow.

3 Main theorem

Theorem 1. Let $h \in L^2(\mathbb{T}, dm)$, let $u = P[h]$, and let $\Omega \subset \mathbb{D}$ be measurable with $\mu(\Omega) = s < \infty$. Then

$$\int_{\Omega} |u(z)|^2 (1 - |z|^2)^{-1} dA(z) \leq 2\pi \log\left(1 + \frac{s}{\pi}\right) \|h\|_2^2. \quad (2)$$

The factor 2 is the least constant which can replace it uniformly for all $s > 0$. Moreover, the corresponding inequality with factor 1 fails for every fixed $s > 0$.

3.1 Upper bound

Proof of (2). Define

$$f(z) = \sum_{n \geq 0} a_n z^n, \quad g(z) = \sum_{n \geq 1} \overline{a_{-n}} z^n.$$

Then $f, g \in H^2(\mathbb{D})$,

$$u(z) = f(z) + \overline{g(z)},$$

and Parseval's identity gives

$$\|f\|_{H^2}^2 + \|g\|_{H^2}^2 = \sum_{n \in \mathbb{Z}} |a_n|^2 = \|h\|_2^2. \quad (3)$$

The source inequality (1), applied to f and g separately, and the pointwise inequality

$$|f + \bar{g}|^2 \leq 2(|f|^2 + |g|^2)$$

give

$$\int_{\Omega} |u|^2 (1 - |z|^2)^{-1} dA \leq 2\pi \log\left(1 + \frac{s}{\pi}\right) (\|f\|_{H^2}^2 + \|g\|_{H^2}^2).$$

Now use (3). □

4 The sharpness family

For $0 < \rho < 1$, let

$$p_{\rho}(e^{it}) = \frac{1 - \rho^2}{|e^{it} - \rho|^2} = \sum_{n \in \mathbb{Z}} \rho^{|n|} e^{int}.$$

Parseval gives

$$\|p_{\rho}\|_2^2 = \sum_{n \in \mathbb{Z}} \rho^{2|n|} = \frac{1 + \rho^2}{1 - \rho^2}.$$

Consequently,

$$h_{\rho} = \sqrt{\frac{1 - \rho^2}{1 + \rho^2}} p_{\rho} \quad (4)$$

has norm one. Its Poisson extension is

$$u_{\rho}(z) = \sqrt{\frac{1 - \rho^2}{1 + \rho^2}} \frac{1 - \rho^2 |z|^2}{|1 - \rho z|^2}. \quad (5)$$

Fix $s > 0$ and set

$$R^2 = \frac{s}{s + \pi}.$$

The disk automorphism

$$\phi_{\rho}(w) = \frac{\rho + w}{1 + \rho w}$$

sends 0 to ρ . Put

$$\Omega_{\rho} = \phi_{\rho}(B(0, R)).$$

Since hyperbolic area is invariant under disk automorphisms,

$$\mu(\Omega_{\rho}) = \mu(B(0, R)) = \frac{\pi R^2}{1 - R^2} = s. \quad (6)$$

4.1 Pullback calculation

Writing $w = x + iy$, direct algebra gives

$$1 - |\phi_\rho(w)|^2 = \frac{(1 - \rho^2)(1 - |w|^2)}{|1 + \rho w|^2}, \quad (7)$$

$$1 - \rho\phi_\rho(w) = \frac{1 - \rho^2}{1 + \rho w}, \quad (8)$$

$$1 - \rho^2|\phi_\rho(w)|^2 = \frac{(1 - \rho^2)(1 + \rho^2 + 2\rho x)}{|1 + \rho w|^2}. \quad (9)$$

Equations (5)–(9) imply

$$(1 - |\phi_\rho(w)|^2)|u_\rho(\phi_\rho(w))|^2 = \frac{(1 - |w|^2)(1 + \rho^2 + 2\rho x)^2}{(1 + \rho^2)|1 + \rho w|^2}. \quad (10)$$

The left side of (10) is the integrand when the desired integral is written against $d\mu$. The closed disk $|w| \leq R < 1$ stays away from -1 , so (10) converges uniformly there as $\rho \uparrow 1$. Invariance of $d\mu$ yields

$$\lim_{\rho \uparrow 1} \int_{\Omega_\rho} |u_\rho(z)|^2 (1 - |z|^2)^{-1} dA(z) = \int_{|w| < R} \frac{2(1 + x)^2}{|1 + w|^2(1 - |w|^2)} dA(w). \quad (11)$$

Lemma 1. For $0 \leq r < 1$,

$$\frac{1}{2\pi} \int_0^{2\pi} \frac{(1 + r \cos \theta)^2}{1 + 2r \cos \theta + r^2} d\theta = 1 - \frac{r^2}{2}.$$

Proof. Put $q = 1 + re^{i\theta}$. Then

$$\frac{(\operatorname{Re} q)^2}{|q|^2} = \frac{1}{2} + \frac{1}{4} \left(\frac{q}{\bar{q}} + \frac{\bar{q}}{q} \right).$$

With $z = e^{i\theta}$, the angular mean of $q/\bar{q}$ is

$$\frac{1}{2\pi i} \int_{|z|=1} \frac{1 + rz}{z + r} dz = 1 - r^2,$$

the residue at $z = -r$. The conjugate has the same mean. □

Using polar coordinates and Lemma 1 in (11),

$$\begin{aligned} \lim_{\rho \uparrow 1} \int_{\Omega_\rho} |u_\rho(z)|^2 (1 - |z|^2)^{-1} dA(z) &= 2\pi \int_0^R \frac{r(2 - r^2)}{1 - r^2} dr \\ &= \pi (R^2 - \log(1 - R^2)) \\ &= \pi \left[\log \left(1 + \frac{s}{\pi} \right) + \frac{s}{s + \pi} \right]. \end{aligned} \quad (12)$$

4.2 Consequences of the limit

The value in (12) exceeds the analytic right-hand side

$$\pi \log \left(1 + \frac{s}{\pi} \right)$$

by $\pi s/(s + \pi) > 0$. Hence, for every fixed $s > 0$, all sufficiently large $\rho < 1$ give actual norm-one counterexamples to factor 1.

If (2) were valid with a uniform factor C in place of 2, (12) would force

$$C \geq 1 + \frac{s/(s + \pi)}{\log(1 + s/\pi)} \quad (s > 0).$$

The right-hand side tends to 2 as $s \downarrow 0$. Thus $C \geq 2$. The upper bound already proves $C = 2$, completing the proof of Theorem 1.

5 Verification and scope

- The constant Fourier mode occurs only in f , so (3) has no double counting.
- The three automorphism identities (7)–(9) were checked by direct expansion.
- A reusable numerical checker evaluates the angular identity at four radii and the pre-limit integral at twelve (s, ρ) pairs. It converges to (12). These checks are not used as proof.
- The theorem answers the literal existence question and determines the exact best constant multiplying the source bound uniformly in s . It does not determine the optimal constant separately for each fixed s , nor classify fixed- s extremizers.

Novelty check. On 23 July 2026, bounded arXiv/web searches used the exact source question, the phrases *Poisson extensions*, *general L^2* , *harmonic concentration*, and the four source authors. They found the source paper but no later exact answer. This is not a comprehensive priority search.

Human review recommendation. Check the source normalization, the pullback density (10), the angular mean, and the distinction between uniform sharpness and fixed- s sharpness. The current verifier verdict is *likely valid*.

References

- [1] J. Gómez, D. Kalaj, P. Melentijević, and J. P. G. Ramos, *Uniform stability of concentration inequalities and applications*, arXiv:2411.16010 (2024). <https://arxiv.org/abs/2411.16010>.